Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Apply Clinical Guideline Rules
2. Abbreviated Title	Apply Guidelines
3. Use Case Id	UC-17
4. Actors	ML Engine (Supporting)
5. Description	The system applies established clinical guideline rules (e.g., ACC/AHA cardiovascular risk thresholds) to provide evidence-based context alongside the ML prediction, enhancing clinician trust and interpretability.
5.1. Pre Conditions	1. Patient feature values are available. 2. Clinical guideline rule set is loaded in the system.
5.2. Task Sequence	1. System evaluates patient data against predefined clinical rules. 2. System flags values exceeding clinical thresholds (e.g., BP > 140). 3. System generates guideline-based recommendations. 4. System appends flags and recommendations to the narrative report.
5.3. Post Conditions	1. Clinical risk flags and guideline-based recommendations are generated. 2. Annotations are ready for inclusion in the diagnostic report.
6. Modification History	Date: September 11, 2026
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